10 Minutes Bebras Practice

You will all be participating in the 2025 Bebras Computational Thinking challenge in the **second week after half term**.

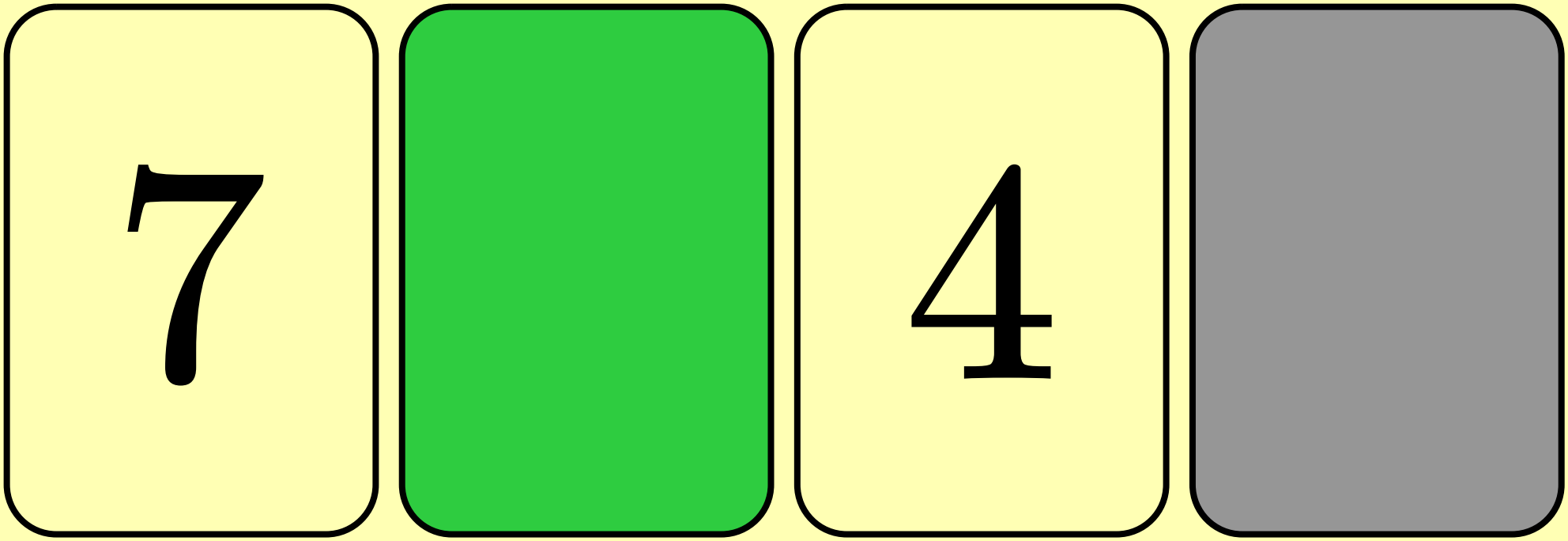
Practice using the **Bebras 2020-2024 Elite challenges**.

Go to **https://bebras.uk** and click on challenge tab.
Click on Elite under 2024 Challenge.

Starter

Which cards do you have to turn over to test the rule:

`If a card has an odd number on one side, then the other side is grey.`



Logic Gates and Boolean Algebra

Keywords:
Logic; Boolean Algebra; Truth Table; AND; OR; NOT; NOR; NAND; XOR; Logic Circuit; Commutative; Associative; Distributive

Logic Gates and Boolean Algebra

Imagine you're a bouncer in spoons. You see four people in front of you...
Who do you need to investigate?

1. Do you need to ask the age of the person drinking a coke?
2. Do you need to ask what the person - who is clearly in their 50s – is drinking?
3. Do you need to ask the age of the person with a pint of lager?
4. Do you need to ask what the person - who is clearly underage - is drinking?

These problems are logically equivalent!

There are some people/cards you need to check because the rule applies directly to them.

These problems are logically equivalent!

There are some people/cards you need to check because the rule applies directly to them.

There are some people/cards you don't need to check as the rule doesn't apply to them.

These problems are logically equivalent!

There are some people/cards you need to check because the rule applies directly to them.

There are some people/cards you don't need to check as the rule doesn't apply to them.

There are some people/cards you need to check because the rule **may** apply to them. *They may break the rule*

Boolean expressions

A **Boolean expression** is a condition that can be TRUE or FALSE

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Boolean expressions can be compounded by means of the operators **AND, OR and NOT**

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Boolean expressions can be compounded by means of the operators **AND, OR and NOT**

Write the truth table and draw out the logic circuit symbols for:

1. OR gate
2. AND gate
3. NOT gate

Practice

Practice

OFF

ON

OFF

ON

ON

Practice

Write a Boolean Expression for:

Practice

Write a Boolean Expression for: $Y = \overline{A}.B$

Practice

Write a Boolean Expression for:

Practice

Write a Boolean Expression for:

$$Y = (A + B) \cdot \overline{\overline{B}}$$

$$= A.\overline{B} + B.\overline{B} = A.\overline{B} + 0 = A.\overline{B}$$

Practice

Write a Boolean Expression for:

$$Y = (A + B) \cdot \overline{\overline{B}}$$

$$= A.\overline{B} + B.\overline{B} = A.\overline{B} + 0 = A.\overline{B}$$

NEVER DO THIS IN AN EXAM

Recall

Write the truth table and draw out the logic circuit symbols for:

1. NOR gate
2. NAND gate
3. XOR gate

Practice

What is the **Boolean Equation** for the following circuit?

Practice

What is the **Boolean Equation** for the following circuit?

$$Y = \overline{A}.B + A.B$$

Practice

$$Y = \overline{A}.B + A.B$$

A	B	Y
0	0	
0	1	
1	0	
1	1	

What is this equivalent to?

Practice

$$Y = \overline{A}.B + A.B$$

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

$$\boxed{Y = B}$$

What is this equivalent to?

Simplifying Boolean Algebra

Why????

How?

Simplifying Boolean Algebra

Why????

Fewer logic gates, means simpler circuits, which means cheaper designs and cheaper processors etc.

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- Using a truth table

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- Using algebra

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- Using a truth table
- Using algebra
- Exam board prefers the algebra method

Simplifying Boolean Algebra

Why????

Fewer logic gates, means simpler circuits, which means cheaper designs and cheaper processors etc.

How?

- Using a truth table
- Using algebra
- Exam board prefers the algebra method
- Maths skills required: factorising and multiplying out brackets

Boolean Algebraic Laws

Commutative

$$A \cdot B = B \cdot A$$

$$A + B = B + A$$

Associative

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C$$

$$A + (B + C) = (A + B) + C$$

Distributive

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

$$A + B \cdot C = (A + B) \cdot (A + C)$$

Simplification using algebra

1. Look at the expression.
2. **Don't panic**
3. Use one of the **identities** or laws of **commutativity, associativity, distributivity** to reduce one part of the expression.
4. **Repeat** step 3 until there is nothing left to reduce.

Example

Let's simplify:

$$A + \overline{A} \cdot A$$

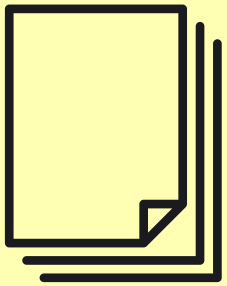
Example

Let's simplify:

$$A + \overline{A} \cdot A$$

$$\Rightarrow A + 0 \qquad \downarrow \overline{A} \cdot A = 0$$

1. $(A + B) \cdot (A + C)$
2. $B \cdot \left(A + \overline{B}\right)$
3. $A + A \cdot B$



Example

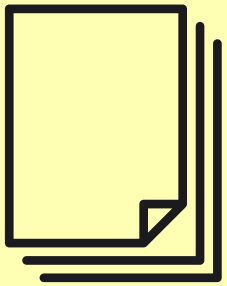
Let's simplify:

$$A + \overline{A} \cdot A$$

$$\Rightarrow A + 0 \qquad \downarrow \overline{A} \cdot A = 0$$

$$\Rightarrow A \qquad \downarrow A + A = 0$$

1. $(A + B) \cdot (A + C)$
2. $B \cdot \left(A + \overline{B}\right)$
3. $A + A \cdot B$



You Try

1. $A \cdot B + \overline{A} \cdot B$
2. $\overline{A} + B \cdot \overline{A}$
3. $\overline{A} \cdot (A + B)$
4. $B \cdot (A + A \cdot B)$
5. $(A + A) \cdot (A + B)$
6. $A + A \cdot A + \overline{A} \cdot B$

You Try

1. $A \cdot B + \overline{A} \cdot B$
2. $\overline{A} + B \cdot \overline{A}$
3. $\overline{A} \cdot (A + B)$
4. $B \cdot (A + A \cdot B)$
5. $(A + A) \cdot (A + B)$
6. $A + A \cdot A + \overline{A} \cdot B$

1. $= B$
2. $= \overline{A}$
3. $= \overline{A} \cdot B$
4. $= A \cdot B$
5. $= A$
6. $= A + B$

More practice!

Google "MIT Boolean Algebra Worksheet"

A `.pdf` titled 'Problem Set 1' should be the top result.

Have a go at the first set of boolean algebra questions, Q1 to Q27

Be aware: They will sometimes skip the $\cdot$ between elements. If there's no symbol, you can assume a dot, eg: $y\overline{y} \equiv y \cdot \overline{y}$